Attempt 4: Final Query Result

The two-dimensional model works locally, but not consistently enough

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1. The Answer First

Final decision

The current shared two-dimensional model is **not** a general replacement for product quantization.

- It has clear wins in some settings.
- It loses or ties in other equally important settings.
- No storage budget passes both datasets and both index settings.

What this means

The implementation works and the idea has a real signal, but the signal is not robust enough for a SIGMOD/VLDB/ICDE contribution in its current form.

2. What Was the Question?

We compress every database vector to either **32 bytes** or **64 bytes**.

Standard approach

Split coordinates into small blocks. Learn a separate set of representative points for every block.

Our approach

Learn one reusable two-dimensional shape. Each coordinate pair stretches, rotates, and shifts that shared shape.

The test

At the same code bytes and the same searched database candidates, does our model recover more true neighbours without slowing queries?

3. How the Method Works

```
Training:
  learn one shared 2D shape
  for each coordinate pair:
    learn a shift + 2-by-2 transform
  store one label per database pair
Query:
  build 64 small tables per searched cell
  scan labels and keep the best 100 IDs
```

Why it might help: a non-rectangular shape can follow natural two-dimensional geometry better than two independent scalar grids.

Why it might fail: every searched cell needs new tables, and one shared shape may be too restrictive for all datasets and rates.

4. Fair Comparison

data	standard SIFT1M and GIST1M queries and exact answers
quality	Recall@100: fraction of the true 100 neighbours recovered
speed	complete queries per second and single-query tail latency
storage	exactly 32 or 64 code bytes per database vector
coarse index	1,024 and 4,096 database cells
search range	nine fixed numbers of cells searched per query
repetition	one warmup and seven measured runs
hardware	one CPU socket; one-thread latency and 12-core throughput

Included in time

Query transformation, cell selection, table construction, database scan, and selection of the best 100 IDs.

5. What Counts as a Real Win?

For one operating point, either:

1. **speed route**: at least 10% more queries per second at nearly the same Recall, without making p95 latency more than 5% worse; or
2. **quality route**: at least 0.002 more Recall at nearly the same throughput.

It must also:

- use no more complete index bytes;
- use no more than twice the baseline construction CPU; and
- not be beaten simultaneously in quality, speed, and bytes by the contextual comparison.

Terminal rule

One byte budget and one route must pass **both datasets and both coarse index sizes**. A single attractive point is not enough.

6. The Complete Matrix Is Valid

Check	Completed	Required
logical method settings	94	94
warmup passes	188	188
measured passes	1,316	1,316
CPU hours	239.25	at most 256
wall hours	101.22	at most 120
peak memory	11.12 GiB	at most 16 GiB

Stability

For every fixed point, all repetitions returned the same IDs, candidate count, and Recall. A second summary run reproduced every output byte-for-byte.

One interrupted warmup remained conservatively charged and was later rerun successfully; it is not a scientific failure.

7. Final Result by Dataset and Index Size

Dataset	Coarse cells	32 bytes	64 bytes
GIST	1,024	**pass**	**pass**
GIST	4,096	**fail: index bytes**	**fail: index bytes**
SIFT	1,024	**fail**	**fail**
SIFT	4,096	**fail**	**pass**

Why the project-level answer is no

Neither column is all green. The 32-byte model loses on SIFT. The 64-byte model helps SIFT only with 4,096 coarse cells, not with 1,024.

8. The Clearest Positive Point

SIFT, 4,096 coarse cells, 64-byte code

Searching 256 cells:

$$\text{Recall} = 0.9261, \qquad \text{throughput} = 966 \text{ queries/s.}$$

Compared with the eligible full-dimensional OPQ point:

- about **56% more throughput** at matched Recall;
- about **29% lower p95 latency**; and
- quality advantage of **0.00214** at matched throughput.

Why this still does not establish the contribution

The benefit appears only at the 64-byte, 4,096-cell, high-search setting. The same model does not repeat it at 1,024 cells or at 32 bytes.

9. The Decisive Negative Evidence Is SIFT

- At 32 bytes and 1,024 cells, S's best raw speed comparison is about **28% slower**; a representative quality comparison is about **0.079 Recall worse**.
- At 32 bytes and 4,096 cells, its best raw speed comparison is about **19% slower**, and the best quality comparison is about **0.042 Recall worse**.
- At 64 bytes and 1,024 cells, quality is nearly tied but slightly negative; S is about **11% slower** with about **17% worse p95 latency**.

Interpretation

The shared shape is useful only after enough code capacity and enough scanned candidates amortize its tables. That is an operating-region effect, not a robust new frontier.

10. Why GIST Looks Better—and Why We Are Careful

GIST has 960 dimensions, but S stores a label only for the first 128 transformed coordinates. For the remaining 832 coordinates, every vector in one coarse cell reuses the cell centre.

more searched cells $\rightarrow$ more approximate tail values $\rightarrow$ possible false positives

What is valid

The GIST numbers are valid for the system definition fixed before seeing queries.

What is not valid

They are not clean proof that the shared 2D shape alone is better. At 4,096 cells, the extra routing state also makes S's complete index larger than the matched baseline.

11. Scientific Result vs. Engineering Result

Scientific evidence

- Shared 2D geometry can improve real query ordering.
- The gain depends strongly on rate and candidate regime.
- Offline reconstruction gains do not predict a common query win.

Artifact engineering

- deterministic packed scanner;
- shared candidate schedule;
- complete byte and resource accounting;
- 101-hour reproducible matrix.

Reviewer view

The engineering makes the negative conclusion trustworthy. It does not turn the local positive points into a general research contribution.

12. Final Takeaway and Next Question

Current direction

Close it under the frozen hypothesis. Do not rescue it by changing datasets, probe grids, thresholds, or storage accounting after seeing results.

Evidence worth preserving

The 64-byte high-candidate SIFT region is real. It shows that the mechanism is not empty; it is simply not robust.

A scientifically different next question

Why does the shared shape help only when code capacity and candidate count are high, and can that mechanism be made robust without a query-selected policy or extra per-cell state?

References and Evidence I

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